

RAMDEOBABA UNIVERSITY



Computer Science & Engineering

PROJECT SYNOPSIS

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Semester: V Semester

Programme: B. Tech in Computer Science and Engineering

(Data Science & Analytics)

ENERGY CONSUMPTION FORECASTING IN SMART GRIDS USING AI MODELS

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1. Background and Motivation

Smart grids integrate digital communication, smart meters, distributed renewable generation, and automated control with conventional power infrastructure, producing vast volumes of real-time consumption data. Accurate forecasting of energy demand is essential for grid stability, cost-effective power generation scheduling, efficient integration of renewable sources, and reduction of transmission and distribution losses. Traditional statistical techniques such as ARIMA and linear regression struggle to capture the non-linear, multi-seasonal, and weather-dependent patterns present in modern smart grid consumption data.

The proposed Energy Consumption Forecasting in Smart Grids platform uses AI-based time-series forecasting models, namely LSTM, GRU, Random Forest, and XGBoost, to predict short-term and medium-term energy consumption at household, feeder, and substation levels. Explainable AI (XAI) techniques such as SHAP accompany every prediction with the underlying contributing factors, while an anomaly-detection and recommendation module converts forecasts into actionable decisions such as load balancing or demand-response alerts, presented through a real-time interactive dashboard.

2. Problem Statement

Existing load forecasting systems used by utilities largely rely on simple statistical models, fail to account for the volatility introduced by renewable generation and changing consumer behaviour, and offer no explanation for their predictions. They do not adjust forecasts in real time as new smart-meter data arrives, provide no concrete load-balancing recommendations once a demand spike is predicted, and are frequently expensive enterprise solutions that remain out of reach for smaller utilities and microgrid operators. Therefore, an explainable, multi-horizon, recommendation-driven AI platform is needed to help

grid operators anticipate demand fluctuations and act on them before they occur.

3. Objectives

Develop a unified platform that forecasts energy consumption at household, feeder, and substation levels across short-term (hourly/daily) and medium-term (weekly/monthly) horizons.

Apply and compare LSTM, GRU, Random Forest, and XGBoost

Integrate Explainable AI (SHAP) so every prediction is accompanied by the factors driving it.

Generate concrete optimisation recommendations (e.g., load shifting, demand-response alerts, peak-shaving schedules) rather than raw predictions alone.

Incorporate external data sources such as weather, temperature, calendar/holiday information, and historical load to improve forecast accuracy.

Build a real-time dashboard showing live consumption, short and medium-term forecasts, confidence intervals, and recommended load-balancing actions.

4. Literature Review and Research Gap

AI and machine learning have been widely applied to power-system load forecasting, but most work addresses either a single forecasting horizon or forecasting accuracy alone, without explainability or multi-node, real-time deployment guidance.

Sr.No.	Author(s) & Year	Title / Focus	Methodology / Key Contribution	Limitation / Gap
1	Kong et al. (2019)	Short-Term Residential	Recurrent architecture applied to smart-meter	Focused on single-household

2	Cho et al. (2014)	Load Forecasting (LSTM)	data to capture temporal load patterns.	residential data; not extended to feeder or substation-scale forecasting.
3	Chen & Guestrin (2016)	XGBoost: A Scalable Tree Boosting System	Gradient-boosted decision trees for high- accuracy structured/ tabular prediction.	Effective for short- term prediction but lacks native explainability and temporal memory.
4	Breiman (2001)	Random Forests	Ensemble of decision trees offering robust, low-variance baseline predictions.	Provides strong baseline accuracy but does not inherently model sequential/ temporal dependencies.
5	Lundberg & Lee (2017)	A Unified Approach to Interpreting Model Predictions (SHAP)	Game-theoretic feature attribution for explaining individual predictions.	Rarely integrated into real-time smart grid forecasting dashboards.

Research Gap: Existing smart grid forecasting systems are largely single-model, purely historical-data-driven, and opaque in how predictions are generated. Very few systems combine multi-horizon forecasting across grid levels, explainable AI, and actionable load-balancing recommendations within a single real-time

platform. The proposed system addresses this gap by unifying forecasting, explainability, and decision support for utilities and microgrids of all sizes.

5. Proposed Methodology / Plan of Work

The proposed system ingests historical smart-meter and substation load records together with external data such as weather, temperature, and calendar/holiday information. After preprocessing and feature engineering, multiple forecasting models – LSTM, GRU, Random Forest, and XGBoost – are trained and compared for each forecasting horizon, and the best-performing model (or an ensemble) is selected per grid node.

Every forecast is passed through an Explainable AI layer using SHAP, producing a ranked list of contributing factors (e.g., a heatwave, a public holiday, a rise in industrial load) alongside the numeric prediction. A recommendation module then translates each forecast and its explanation into a specific action – such as shifting flexible loads, triggering demand-response alerts, or scheduling additional generation – which is surfaced on a real-time dashboard along with live consumption, short-term forecasts, confidence intervals, and anomaly alerts.

Data ingestion – smart-meter and substation records + external sources (weather, calendar, holidays)

Preprocessing & feature engineering – cleaning, merging, temporal feature creation

Multi-model forecasting – LSTM, GRU, XGBoost

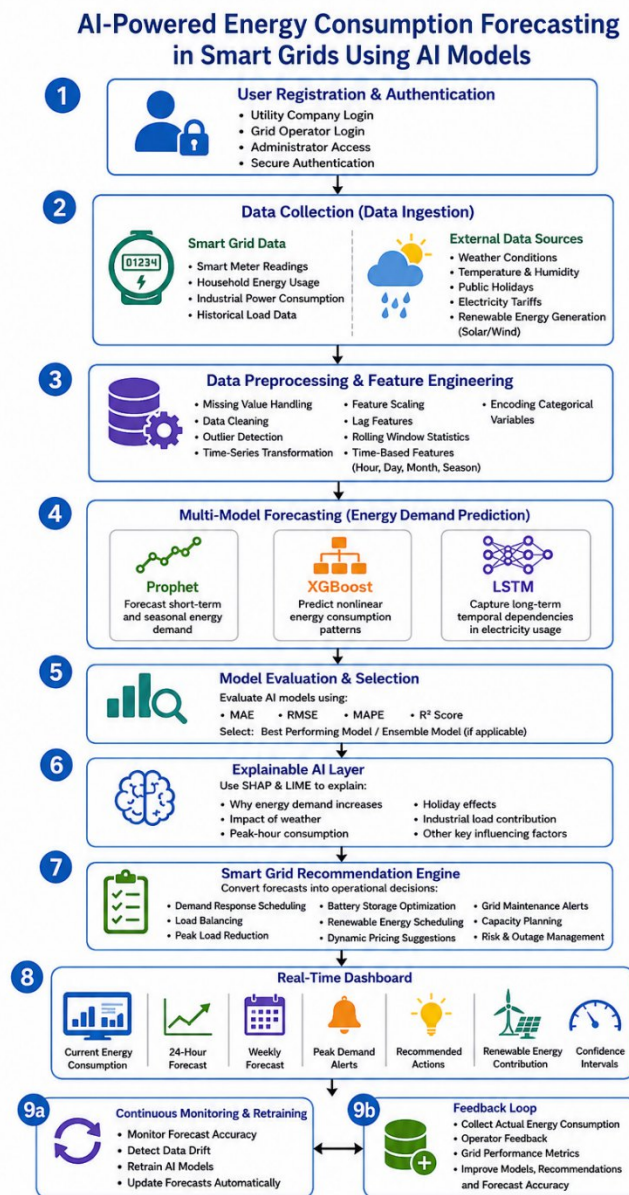
Model comparison & selection – accuracy evaluation across grid nodes

Explainability layer – SHAP attribution for each prediction

Recommendation engine – converts forecasts + explanations into actionable steps

Real-time dashboard – live consumption, forecasts, alerts, and recommended actions

Continuous monitoring & retraining – models updated as new smart-meter data arrives



6. Technology, Tools and Platforms

The proposed system will be developed using Python and React.js with FastAPI as the backend framework. PostgreSQL and a time-series database (InfluxDB) will be used for data management, while the forecasting models will be implemented using PyTorch/TensorFlow (LSTM, GRU), scikit-learn (Random Forest), and XGBoost. Explainable AI will be implemented using the SHAP library. Docker and GitHub will support deployment and version control, and an MQTT-based ingestion service will stream real-time smart-meter data. Load and weather datasets will be sourced from smart-meter records, utility SCADA systems, meteorological department data, and public energy-consumption datasets.

7. Expected Outcomes, Deliverables and Functional Specifications

The proposed project will deliver an AI-powered energy consumption forecasting system that predicts demand across household, feeder, and substation levels for short-term and medium-term horizons. The system will provide explainable predictions, actionable load-balancing recommendations, and a real-time interactive dashboard with anomaly alerts and confidence intervals. Key functionalities include multi-horizon forecasting, model comparison across LSTM/GRU/Random Forest/XGBoost, SHAP-based explanations, a recommendation engine, and integration of external contextual data. The final deliverables will include a fully functional web application, trained forecasting models, a secure time-series database, system documentation, testing reports, and a research paper evaluating forecasting accuracy, interpretability, and practical applicability.

8. Project Scope

The proposed project develops an AI-powered web-based energy consumption

forecasting platform focused on forecasting, explainability, and load-balancing recommendations across multiple grid levels. It supports multi-horizon demand prediction, explainable AI-based reasoning, real-time alerts, and decision-support recommendations. The system focuses on forecasting and recommendation support only, excluding automated control of grid switching equipment, direct tariff-setting, and legally binding regulatory decisions.

9. Project Timeline

The project will be completed over 16 weeks, covering literature review, requirement analysis, system design, model and application development, testing, performance evaluation, documentation, research paper preparation, and final project submission.

Phase	Duration	Milestone / Deliverable
Literature review & requirement analysis	Week 1 – Week 3	Problem identification, literature survey, requirement gathering, project planning
Design & methodology finalisation	Week 4 – Week 5	System architecture, data pipeline design, database design, UI/UX design, XAI framework design
Implementation / development	Week 6 – Week 12	Data collection & preprocessing, forecasting model development (LSTM, GRU, Random Forest, XGBoost), SHAP integration, recommendation engine, dashboard development
Testing & evaluation	Week 13 – Week 14	Model accuracy evaluation, explainability validation, functional testing, bug fixing, performance optimisation
Documentation & final submission	Week 15 – Week 16	Project report, research paper, user manual, presentation, and final project submission

10. References

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Group Member Details

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